

Tobias Großbölting

Ph.D. Candidate in Economics, University of Mannheim

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Citizenship: German

Placement Director: Prof. Christoph Rothe

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RESEARCH FIELDS

Econometrics

Regression discontinuity designs, nonparametric estimation, bias-aware inference

DOCTORAL STUDIES

University of Mannheim, CDSE

September 2021 – June 2027 (Expected)

Ph.D. Candidate in Economics

Supervisors: [Prof. Christoph Rothe](#) and [Prof. Paul Goldsmith-Pinkham](#)

Yale University

2024 – 2025

Visiting Ph.D. Student.

PRE-DOCTORAL STUDIES

Humboldt University of Berlin

2018 – 2021

M.Sc. Economics.

University of Copenhagen

2019

Visiting Student, Department of Economics.

Humboldt University of Berlin

2015 – 2018

B.Sc. Economics.

University of Warwick

2017 – 2018

Visiting Student, Department of Economics.

REFERENCES

Christoph Rothe

Professor of Economics

University of Mannheim

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Paul Goldsmith-Pinkham

Associate Professor of Finance

Yale School of Management

paul.goldsmith-pinkham@yale.edu

Soonwoo Kwon

Assistant Professor of Economics

Brown University

soonwoo_kwon@brown.edu

JOB MARKET PAPER

Heterogeneous Groups and Covariate Adjustment in Regression Discontinuity Designs

I study covariate adjustment in regression discontinuity designs with discrete covariates, such as education level, occupation, or gender, that partition the sample into subgroups with heterogeneous regression functions. The standard local-polynomial RD estimator pools observations across these heterogeneous groups, generating an additive pooling bias whenever group composition varies along the running variable. I show that covariate adjustment can eliminate this pooling bias from the leading term of the worst-case bias under a uniform asymptotic framework. Specifically, I propose the Stratified RD estimator, which avoids pooling through stratification: It first estimates group-specific treatment effects within each group and group weights at the cutoff and then forms a weighted average. I derive an explicit range of bandwidth choices under which the Stratified RD estimator achieves both weakly lower asymptotic variance and worst-case bias than the local-polynomial estimator, over a common smoothness class. These improvements yield shorter honest confidence intervals for the same target

parameter. A placebo exercise using data from the Current Population Survey shows reductions in MSE of 8% to 33% and honest confidence intervals that are 6% to 16% shorter depending on sample sizes and bandwidths.

WORK IN PROGRESS

Spatial Regression Discontinuity Designs

joint with Soonwoo Kwon, Michael Pollmann and Aurel Rochell

TEACHING EXPERIENCE

University of Mannheim

Econometrics I (PhD-level), Prof. Christoph Rothe
Introduction to Econometrics, Prof. Carsten Trenkler
Statistics II, Dr. Toni Stocker

Fall 2025
Spring 2023, 2024
Fall 2022, 2026

Humboldt University of Berlin

Macroeconomics I, Prof. Lutz Weinke
Macroeconomics II, Prof. Michael Burda
Macroeconomics I, Prof. Michael Burda

Fall 2020
Spring 2020
Fall 2019

CONFERENCE AND SEMINAR PRESENTATIONS

2026

NASM of the Econometric Society, Emory University; ENTER Jamboree, UC3M; ENTER Seminar, UCL; HKMetrics Workshop, Karlsruhe Institute of Technology

2025

CIML Training Conference: Frontiers of Causal Inference and Machine Learning, IMT Lucca; Econometrics Lunch, Brown University; Econometrics Prospectus Lunch, Yale University

2024

Bonn-Mannheim-Munich Workshop in Microeconometrics, University of Mannheim; Workshop on Networks and Political Economy, Paris 1 Panthéon-Sorbonne; CDSE Seminar, University of Mannheim; Econometrics Research Seminar, University of Mannheim; Econometrics Prospectus Lunch, Yale University

2023

CDSE Seminar, University of Mannheim; Econometrics Research Seminar, University of Mannheim

2022

FHM+ Workshop on Development Economics, Heidelberg University

SUMMER SCHOOLS

Gerzensee Summer School: ML for Treatment Effects & Structural Equation Models

Prof. Victor Chernozhukov

2024

Northwestern Advanced Causal Inference Workshop

Prof. Peter Hull, Prof. Christian Hansen, Prof. Andrew Goodman-Bacon

2024

CEMFI Summer School: Causal Inference for Medical and Social Science

Prof. Miguel Hernán

2023

OTHER PROFESSIONAL EXPERIENCE

Research Assistance

Prof. Christoph Rothe, University of Mannheim

2023 – 2024

Prof. Michael Burda, Humboldt University of Berlin

2020 – 2021

Prof. Alexander Stomper, Humboldt University of Berlin

2018 – 2019

AWARDS AND DISTINCTIONS

CDSE Scholarship – University of Mannheim	<i>2026 – 2027</i>
CDSE Scholarship – University of Mannheim	<i>2021 – 2023</i>
Studienförderung der Friedrich-Ebert-Stiftung	<i>2017 – 2021</i>
Government-funded scholarship granted for academic excellence and social engagement.	
Portfolio Management Program – P & K Pühringer Foundation, ESMT and HU Berlin	<i>2018 – 2020</i>
Group-based active portfolio management; certificate and grant for excellent performance. Mentor: Dr. Gerald Leitner.	

MISCELLANEOUS

Programming	R, Python, Java
Software	LaTeX, Zotero
Languages	German (native), English (fluent), Spanish and Italian (beginner)